

Daniel Brethauer

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EDUCATION

Northwestern University 2017-2021

Double Major: Integrated Science Program and Physics

Dean's List Quarters Spring 2021, Winter 2021, Spring 2018, Spring 2019, and Fall 2019

University of California, Berkeley Fall 2021-Present

Astrophysics PhD Candidate

PUBLICATIONS

Impact of Systematic Modeling Uncertainties on Kilonova Property Estimation - First Author

Quantifying the impact of uncertainties in atomic data and radioactive thermalization efficiency on the ability to recover kilonova ejecta properties (Brethauer et al. 2024 ApJ 975 213)

Seven Years of Coordinated Chandra-NuSTAR Observations of SN2014C unfold the Extreme Mass-loss History of its Stellar Progenitor – First Author

Updates to previous work on SN2014C and presents a collection of 121 interacting SNe with parameters of the material they interacted with in a machine readable format. (Brethauer et al. 2022 ApJ 939 105)

Research Notes: Late-Time Chandra-NuSTAR Observations of SN2014C - First Author

Full Title of the Research Note: *Six years of luminous X-ray emission from the strongly interacting type-Ib SN 2014C captured by Chandra and NuSTAR* (Brethauer et al. 2020 Res. Notes AAS 4 235)

SUCCESSFUL PROPOSALS

Late-Time Near/Mid-Infrared Nebular Spectroscopy of a Kilonova with JWST (\$159k)

3 epochs of near- and mid-infrared spectroscopy of a kilonova at late times to constrain the presence of 3rd peak r-process elements and their atomic transitions. JWST Cycle 4-5 (Summer 2025-2027).

Mapping extreme mass loss from evolved massive stars with coordinated Chandra-NuSTAR observations

Coordinated hard and soft X-ray observations of a strongly interacting supernova to understand the last decades to centuries of the star's life prior to explosion through eruptive mass-loss. Chandra Cycle 24

EXPERIENCE

Programming Experience

Language/software proficiencies: Python (numpy, matplotlib, pandas, scipy), C++, Sedona, SQL, Lua, OpenMP, MPI, X-SPEC, CIAO, XSELECT, NuSTARDAS, XIMAGE

Co-Author Publications

AT2018cow – Margutti et al. 2019, ApJ 872 18

The Young Supernova Experiment – Jones et al. 2021 ApJ 908 143

AT2020xnd - ATel #14156 and ATel #14154, Bright et al. 2022, ApJ 926 112

SN 2004C – DeMarchi et al. 2022 ApJ 928 84

Survival of the Fittest: Numerical Modeling of SN 2014C–Vargas et al. 2022, ApJ 930 150

AT2018hyz - Cendes et al. 2022, ApJ 938 28

ToO Observations of GW Events with Vera C. Rubin Observatory-Andreoni et al. 2022, ApJ 260 18

GRB221009A – GCN 32695, 32788

AT2022tsd – AstroNote 2022-218, 2022-229, 2022-237

Graduate Student Instructor at UC Berkeley

Astro C10 – Introduction to Astronomy, for non-majors – Spring 2022

Astro 160 – Stellar Physics – Fall 2022

Head Graduate Student Instructor Astro C10 – Introduction to Astronomy, for non-majors – Fall 2023

MPS 375 - Professional Preparation: Supervised Teaching in Math and the Physical - Fall 2024

Software Sustainability Institute Intermediate Python Course Participant	Summer 2022
An intensive Python course for teaching skills to develop and release software packages	
Zwicky Transient Facility Summer School Participant	Summer 2022
An intensive course run by the Zwicky Transient Facility to teach astronomy-based data science skills	
Celebrating 20 Years of Swift Discoveries Conference Presenter	Spring 2025
Presented my work on quantifying systematic modeling uncertainties of kilonova and the crucial impact that the Swift UVOT telescope can have in making progress for kilonova science	
Rise_Time 2024 Presenter	Summer 2024
Presented my work on quantifying systematic modeling uncertainties of kilonova	
Hot-Wiring the Transient Universe Conference	Summer 2019
Presented my research on SN2014C as detailed below by creating a poster and discussing it with graduate students, post-docs, and professors from across the world.	
Chandra Frontiers in Time-Domain Science Presenter	Fall 2020
Talk on current research and X-ray modeling of SN2014C (https://www.youtube.com/watch?v=BWWU2navp6E , starting at 38:48)	
Supernova/Dust Tele-Talk Series	Summer 2022
Presented my research on my upcoming published work on SN2014C (Brethauer et al. 2022 ApJ 939 105)	
Northwestern Undergraduate Research Peer Mentor	Spring 2019 - Fall 2019
Served as a role model for a group of 8-12 mentees in navigating the challenges of undergraduate research, ranging from writing grant proposals to problem solving during research.	

AWARDS

Northwestern Weinberg Summer Research Grant (\$3.5k)	Summer 2018
NASA Illinois Space Research Grant (\$5k)	Spring 2019
NASA Illinois Space Research Grant (\$8k)	Spring 2020
AAS 237 Chambliss Astronomy Achievement Student Award Medalist	Winter 2021
National Science Foundation Graduate Research Fellowship Program Honorary Mention	Spring 2022
Hearts to Humanity Eternal Grant (\$10k)	Fall 2022
NASA Future Investigators in NASA Earth and Space Science and Technology (FINESST) (\$150k)	
Project: Using Machine Learning to Augment and Explore the Prohibitively Large Parameter Space of Emission from Kilonova. Active from Fall 2023 to Fall 2026.	

ACTIVITIES

UC Berkeley Teaching Certificate Program	Fall 2021 - Present
Hearts to Humanity Eternal Guardian	Fall 2022-Present
Wonderfest Science Envoy	Fall 2024 - Present
Participant in a workshop series to improve science communication skills and reach the broader Bay Area community such as public lectures and high school outreach.	

VOLUNTEER/OUTREACH

Letters to a Prescientist Pen Pal	Fall 2019 - Present
Engage with elementary school children via letters about being a scientist and inspiring scientific curiosity	
Popping the Science Bubble Presenter (Organizing Committee)	Fall 2021 (Fall 2022 – Present)
Present complex scientific research to the general public in an accessible manner to build scientific literacy	
SPLASH Teaching Day	Spring 2022 - Present
Taught introductory astronomy at Splash, a program designed to teach high schoolers	
Astro Nights Organizer	Fall 2021 - Present
Public outreach coordinator, organize public lectures and stargazing events at UC Berkeley once a month. Presented in Spring 2024 (https://www.youtube.com/watch?v=O2lR68OWL3w)	